Yixing Zhao

Curriculum Vitae

Department of Statistics and Actuarial Science, University of Waterloo
Waterloo, Ontario, Canada

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Research Interests

Bayesian nonparametric inference; survival analysis under complex censoring; conditional distribution estimation and machine learning; mortality forecasting.

Education

University of Waterloo, Waterloo, Ontario, Canada

Ph.D. in Statistics (Biostatistics) Sep 2024 - Present
Supervisor: Prof. Liqun Diao. **Expected completion: 2028.**

Master of Mathematics in Statistics Sep 2023 - Aug 2024

Bachelor of Mathematics in Computational Mathematics Sep 2020 - Jun 2023
Minors: Statistics; Combinatorics and Optimization; Computing.

Methodological Research

MAP-EM Pólya Trees for Survival Analysis with Complex Censoring

- Developed a unified Bayesian nonparametric framework using a Pólya tree prior for right-censored, current-status, and interval-censored event-time data.
- Derived closed-form maximum a posteriori expectation-maximization (MAP-EM) updates; studied the estimator's asymptotic properties.
- Evaluated estimation accuracy in simulation studies: sum of squared errors (SSE) was approximately 50% of that of the nonparametric maximum likelihood estimator and 15% lower than that of fiducial methods.

Neural Regression Trees for Censored and Uncensored Outcomes

- Developing conditional distribution estimators using a fixed binary-tree structure with covariate-dependent split locations and branch probabilities determined by a neural network.
- The construction yields conditional density, distribution, and survival functions for exact and censored continuous outcomes.
- Research includes validity of the distribution construction and uniform approximation as tree and network complexity increase.

Conference Presentations

Zhao, Y., and Diao, L. "Neural Regression Trees for Censored and Uncensored Outcomes." Waterloo Student Conference in Statistics, Actuarial Science and Finance, University of Waterloo. October 23, 2026.

Zhao, Y., and Diao, L. "MAP-EM Polya Trees for Survival Analysis with Complex Censoring." 2026 SSC Annual Meeting, Hamilton, Ontario, Canada. June 16, 2026.

Applied Research Experience

Research Intern, Mitacs Accelerate May 2025 - Present

- Developed post-COVID-19 forecasts from historical cause-specific mortality data using DSA variable selection, BMBE modeling, and age-segment selection. Validation SSE was approximately 30% lower than the Lee-Carter benchmark.
- Implemented **mortselect** for reproducible data processing, model fitting, validation, and prediction.
- Built a Bayesian MCMC model using historical mortality shock patterns from World War II, HIV/AIDS, and the Spanish flu for post-pandemic scenario forecasting.

Research Assistant - Mortality Forecasting Winter 2024

Teaching Experience

Graduate Teaching Assistant, University of Waterloo Sep 2023 - Present
Provided tutorials, student support, and assessment across nine courses.

Course	**Terms**
STAT 431: Generalized Linear Models and their Applications	Winter 2026; Fall 2026
STAT 331: Applied Linear Models	Spring 2026
STAT 231: Statistics	Winter 2026; Spring 2026
STAT 221: Statistics (Non-Specialist Level)	Fall 2025
STAT 332: Sampling and Experimental Design	Fall 2024; Spring 2025
STAT 230: Probability	Spring 2025
STAT 437: Statistical Methods for Life History Analysis	Winter 2025
ME 202: Statistics for Engineers	Winter 2025
STAT 333: Stochastic Processes 1	Fall 2024

Academic Service and Mentoring

Mentor, Women in Mathematics Directed Reading Program Winter 2025; Spring 2026
University of Waterloo
Guided students in reading research papers and textbooks on Bayesian analysis and discussing applications in medicine and finance.

Scholarships and Awards

University of Waterloo Graduate Scholarship	2025, 2026
Women in Mathematics Mentorship Award	2025, 2026
Statistics & Actuarial Science Chair's Award	2024, 2026
International Master's Award of Excellence	2024

Research Software & Statistical Computing

mortselect | R package and source code | Interactive web interface
Zhao, Y. (author and maintainer), Diao, L., Meng, Y., and Weng, C.
R package implementing DSA, age-specific band selection, and BMBE forecasts. Developed a browser-based explorer for saved forecasts, with interactive charts and CSV/SVG downloads.
Tools: R, Python, SAS, C++, Git/GitHub.